

MySQL best when designing a small web based with a small volume data, example, when building a local eCommerce store. It use OLAP/OLTP systems as they don't require complex queries and large volumes data.

Oracle widely used in financial sector, where data integrity and security are paramount, they rely on Oracle to manage sensitive financial data and transactions.

PostgreSQL it is perfect for data analysis and warehouse due to complicated queries and wide choice of custom interfaces accomplished with predefined functions.

MongoDB works best in real time data integration and database scalability. Its the right option for product catalogs due to its capacity to stock a multiplicity of objects with various attribute collections.

- a. ETL which stands for Extract, Transform and Load, is the foundation of data engineering. It involves the normalization and consolidation of data collected from many sources for analysis and reporting. It entails gathering data in its most basic form from many sources, cleaning and organizing it, and then feeding it into a target database or data warehouse. ETL is essential for the integration, qualification and facilitation of data access, all of which are vital for corporate decision making. Obtaining raw data from various source systems, organizing, cleaning consolidating, & assembling data are the 3 main tasks.